

Multiple Hypothesis Testing (in DEG, GWAS)

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An example: Genome-wide Association Studies

GWAS

- Input:

- A genotype matrix X ($n \times p$), where $X_{ij} \in \{0, 1, 2\}$
- A phenotype vector $\mathbf{y}$ ($n \times 1$)

- Optimization:

$$\mathbf{y} = \mathbf{x}_j \beta_j + \epsilon, \text{ for each } j$$

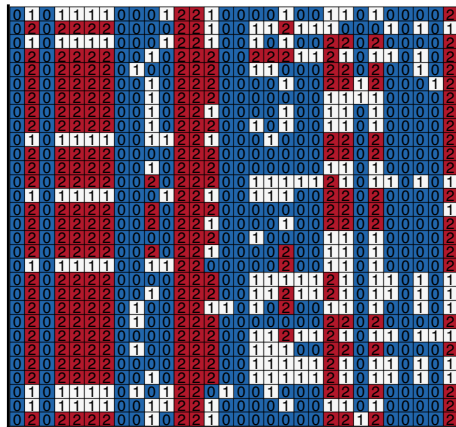
- Output:

- Estimate an effect size (coefficient) β_j ($j \in 1, \dots, p$)

- Testing $H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$.

GWAS to map locations associated with case vs. control tests

A genotype matrix X ($X_{ij} \in \{0, 1, 2\}$):



A vector of phenotypes Y ($Y_i = 1$ if case vs. $Y_i = 0$ if control)

```
table(Y)
```

```
## Y
```

```
##      0      1
```

```
## 1272 1218
```

Goal: Find locations (columns of X) where a genotype x_j is associated with a phenotype y

GWAS by variant-by-variant association tests

A variant-by-variant association t-test for a variant j :

$$H_0 : \mathbb{E}[X_{ij}|Y_i = 1] = \mathbb{E}[X_{ij}|Y_i = 0] \quad \text{vs.}$$

$$\mathbb{E}[X_{ij}|Y_i = 1] \neq \mathbb{E}[X_{ij}|Y_i = 0]$$

- The average genotype is the same between the case and control under the null.

```
## library(matrixTests)
.gwas.file <- "../data/genotype/gwas_results.rds"
if.needed(.gwas.file, {
  .gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ])
  saveRDS(.gwas, .gwas.file)
})
```

Today's lecture

- 1 Hypothesis testing in terms of two errors
- 2 Controlling Family-Wise Error Rate
- 3 False Discovery Rate
- 4 FWER, FDR, then what?
- 5 Beyond global FDR: gene-by-gene localized control

Recap: Steps of hypothesis testing

① Choose a test statistic and formulate hypotheses

- Null hypothesis H_0 : no effect / no difference

② Define the rejection region

- Set significance level α (e.g., 0.05)
- Determine critical values or thresholds

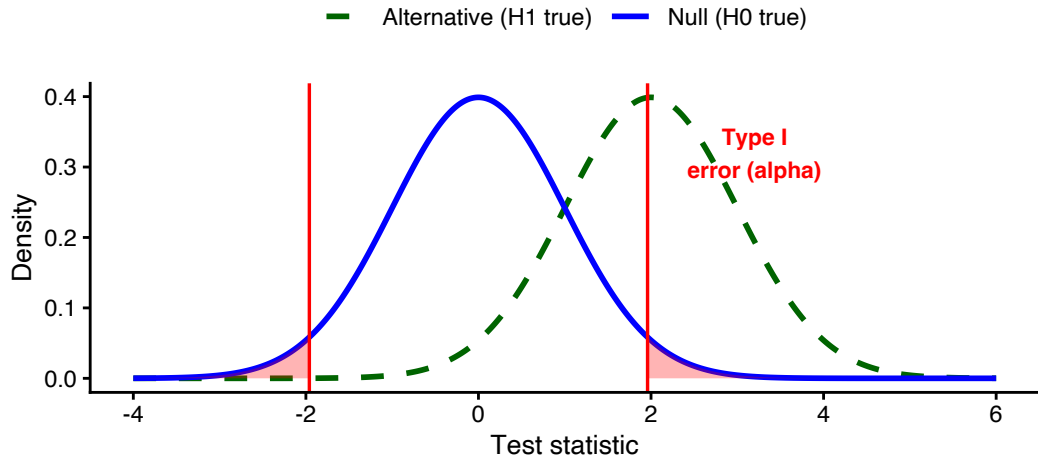
③ Compute the test statistic from observed data

④ Make a decision

- If test statistic falls in rejection region (or p-value $< \alpha$): reject H_0

What does type I error mean for a single test?

Type I error: rejecting H_0 when H_0 is true



Type I and Type II errors (lect04)

	H_0 True	H_0 False
Reject H_0	Type I Error (α)	Correct (Power)
Fail to reject	Correct	Type II Error (β)

- **Type I Error** (false positive): Reject H_0 when it is actually true
- **Type II Error** (false negative): Fail to reject H_0 when it is actually false

Statistical power: $1 - \beta$

Power = Probability of correctly rejecting H_0 when it is false

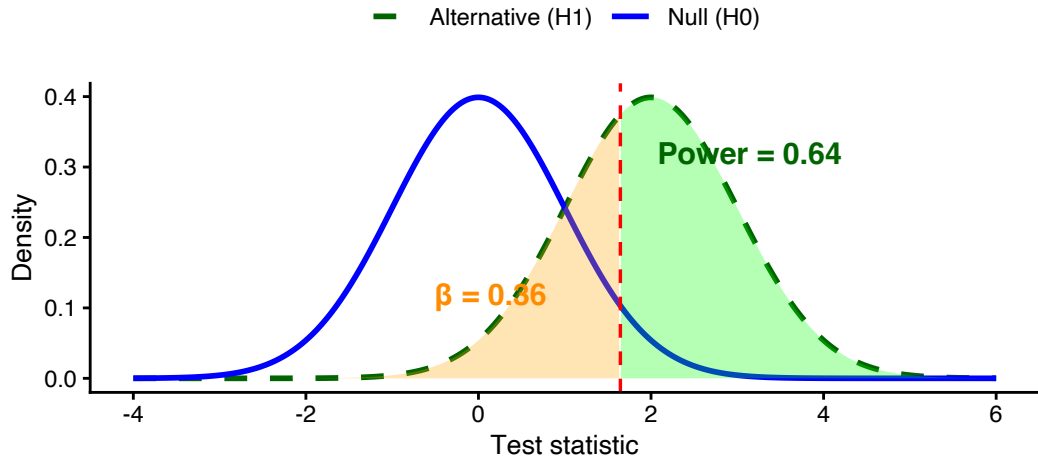
$$\text{Power} = 1 - \beta = P(\text{reject } H_0 \mid H_1 \text{ is true})$$

Power depends on:

- 1 **Significance level** α (Type I error rate)
- 2 **Effect size** δ (how different H_1 is from H_0)
- 3 **Sample size** n
- 4 **Variability** σ

Visualizing power

Power analysis: $\alpha = 0.05$, effect size = 2.0 SD



Why statistical power changes with n

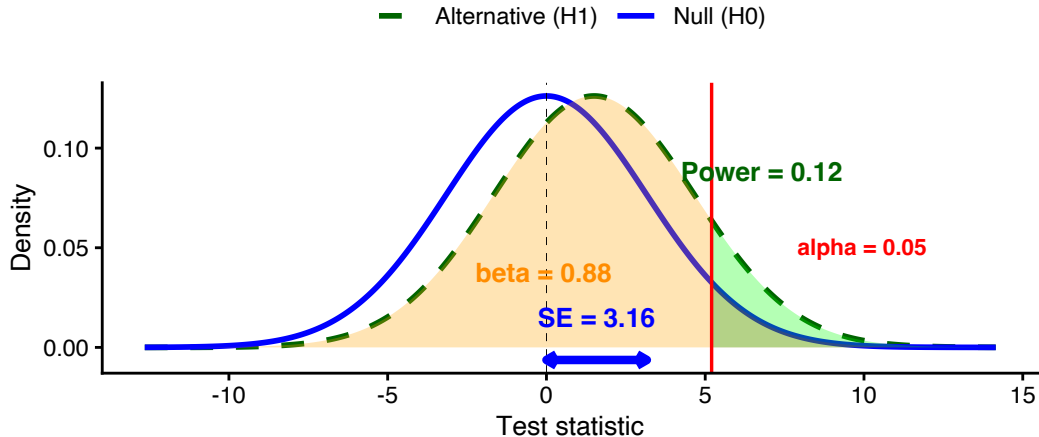
Key insight: As $n \uparrow$, the standard error $\downarrow$

$$SE = \frac{\sigma}{\sqrt{n}}$$

- **Both** null and alternative distributions have the same SE
- Larger $n \rightarrow$ **both distributions become narrower**
- Fixed effect size $\delta \rightarrow$ distributions **separate more** (relative to SE)
- Less overlap $\rightarrow$ easier to distinguish H_0 from H_1
- Result: **higher power**

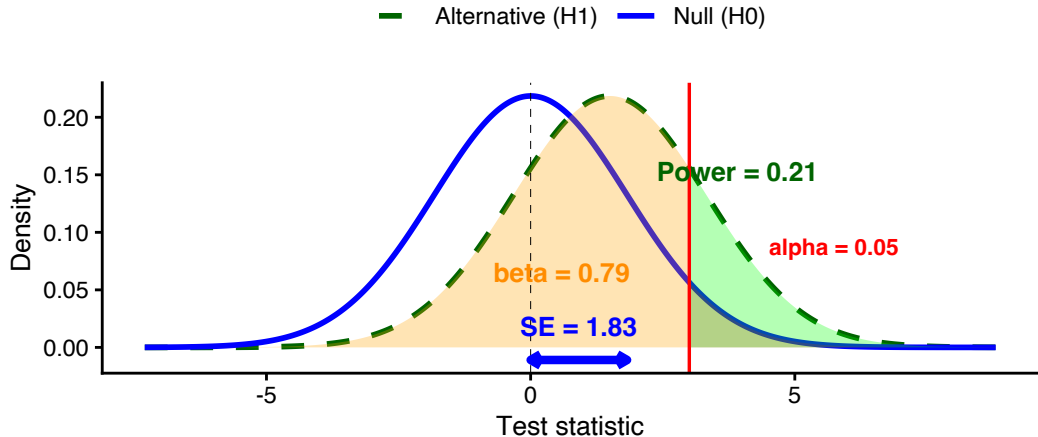
Power increases with sample size: visualization

$n = 10$



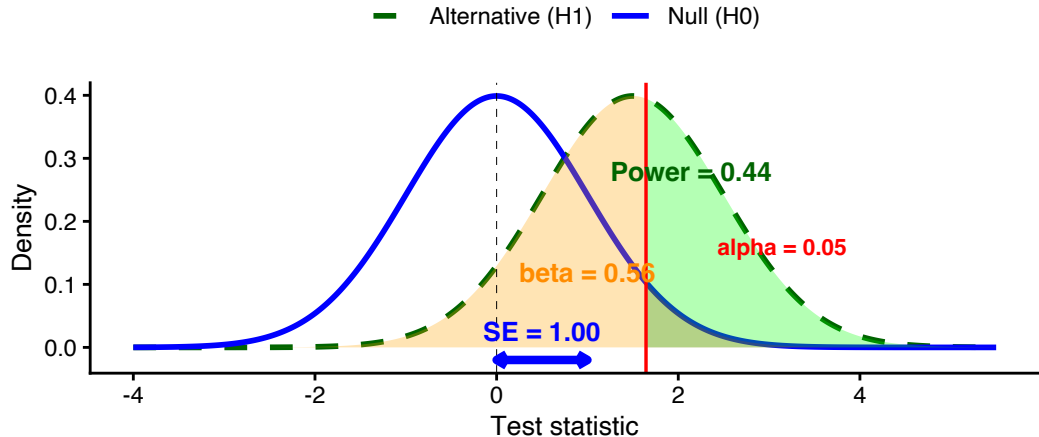
Power increases with sample size: visualization

$n = 30$



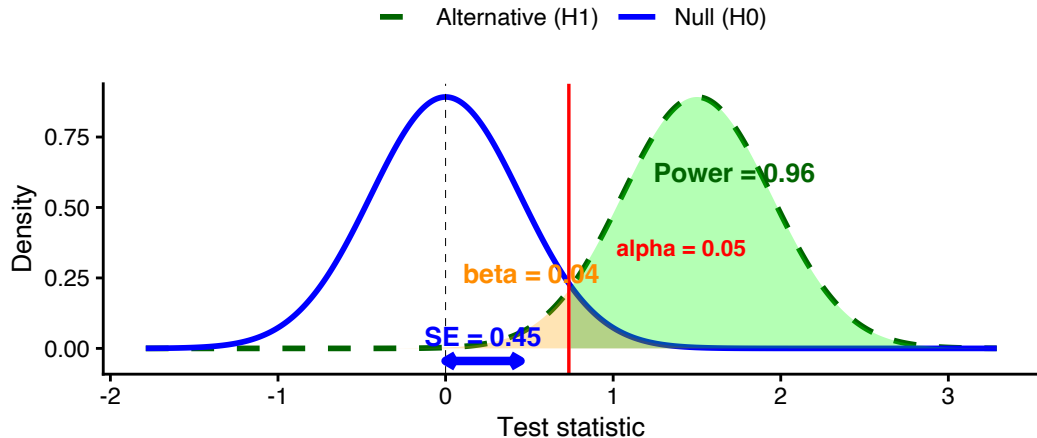
Power increases with sample size: visualization

$n = 100$



Power increases with sample size: visualization

$n = 500$



What do we need to perform power calculation?

① Significance level (α)

- Type I error rate we're willing to accept

② Effect size (δ or Cohen's d)

- Can estimate from pilot data or literature

③ Sample size (n)

- Number of observations per group

④ Standard deviation (σ)

- Estimate from prior experiments

⑤ Power ($1 - \beta$)

- Higher power = more confident in detecting true effects

Typical questions:

- Given α , δ , σ , and desired power $\rightarrow$ find n
- Given α , n , σ , $\delta \rightarrow$ find power

Power calculation in R (e.g., t-test)

Question: How much power do we have with the sample size n ?

```
power.t.test(  
  delta = 0.5,      # Effect size (mean diff)  
  sd = 1,           # Standard deviation  
  n = 50,           # Sample size per group  
  sig.level = 0.05, # Rejection region (desired Type I)  
  type = "two.sample"  
)
```

Answer: roughly 0.7 power

Sample size calculation

Question: How many samples do we need to achieve 80% power?

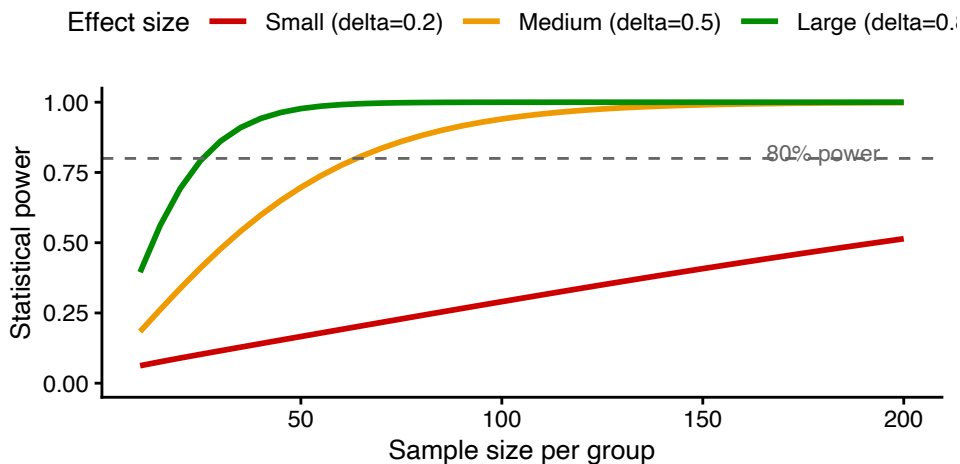
Calculate required sample size for 80% power

```
power.t.test(  
  delta = 0.5,      # Effect size...  
  sd = 1,           # Standard deviation  
  power = 0.8,      # Desired power  
  sig.level = 0.05,  
  type = "two.sample"  
)
```

Answer: ≈ 64 samples per group for the effect size $\delta = 0.5$

Power curves: sample size vs. power

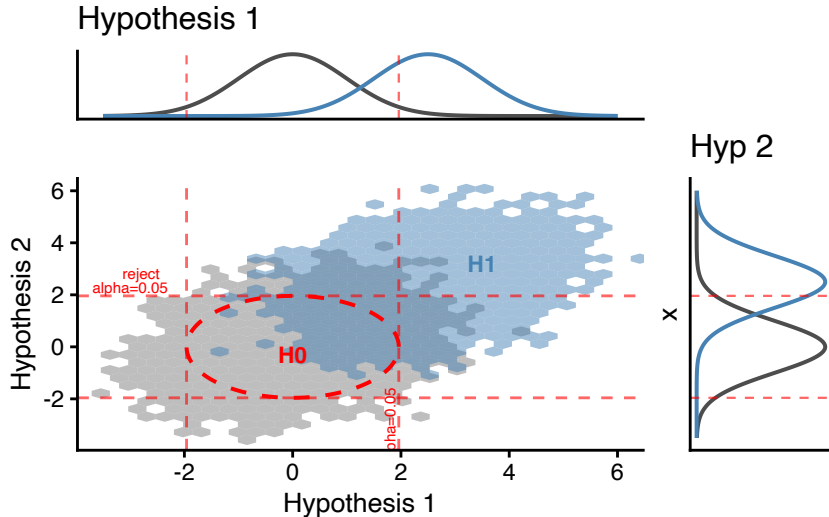
Power curves for two-sample t-test ($\alpha = 0.05$)



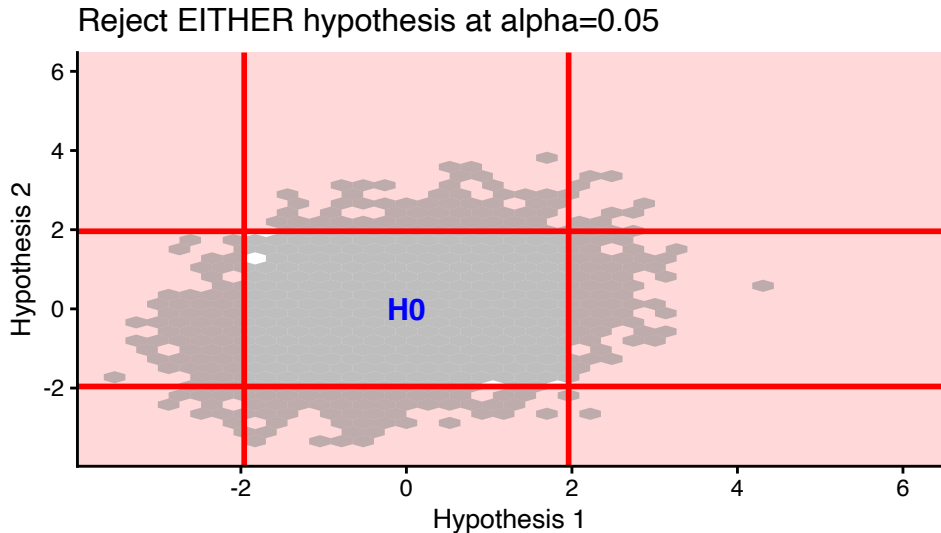
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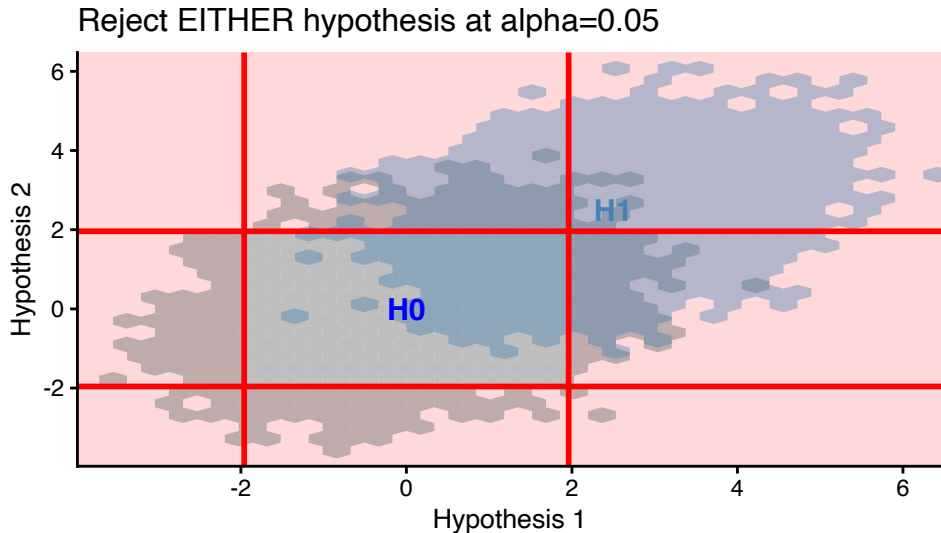
What if we have more than one hypothesis?



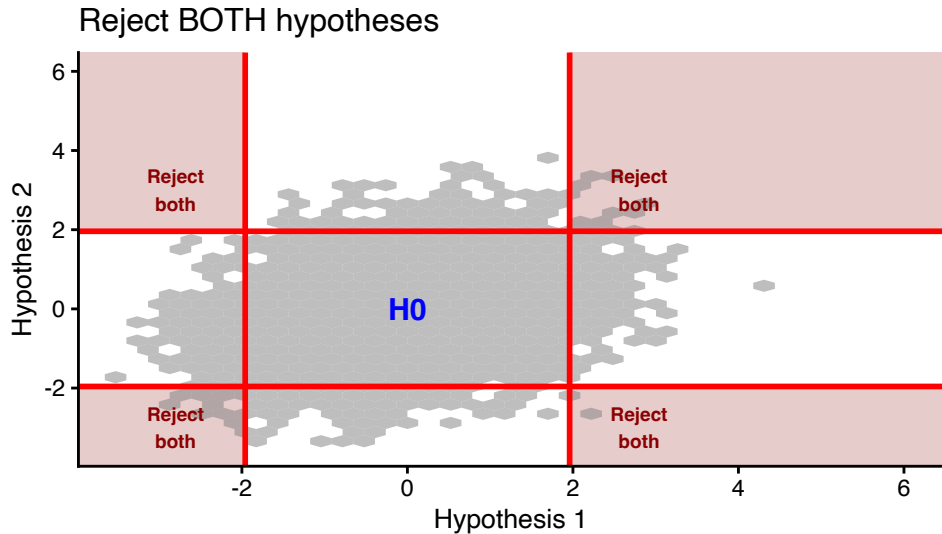
What if we decided to reject either one at $\alpha < 0.05$



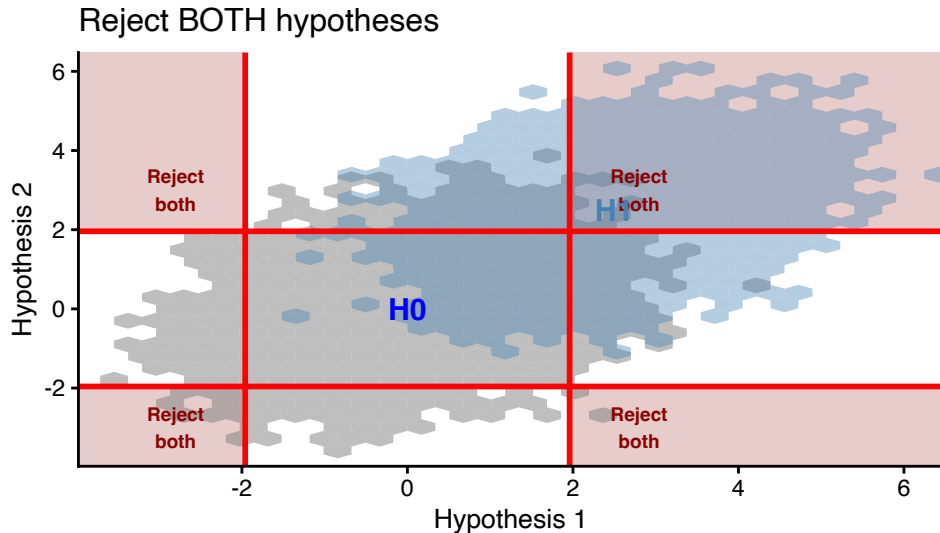
What if we decided to reject either one at $\alpha < 0.05$



What if we test each hypothesis $\alpha < 0.05$?



What if we test each hypothesis $\alpha < 0.05$?



Wait, what was the definition of p-value?

P-value: Probability of observing a test statistic as extreme or more extreme than what we observed, **assuming the null hypothesis is true**

$$P(\text{data as extreme or more} \mid H_0 \text{ is true})$$

For a single test at $\alpha = 0.05$:

- $P(\text{Type I error}) = P(\text{reject } H_0 \mid H_0 \text{ is true}) < \alpha = 0.05$

What about two p-values?

Union rule (probability of A or B):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) < P(A) + P(B)$$

Independence (probability of A and B when independent):

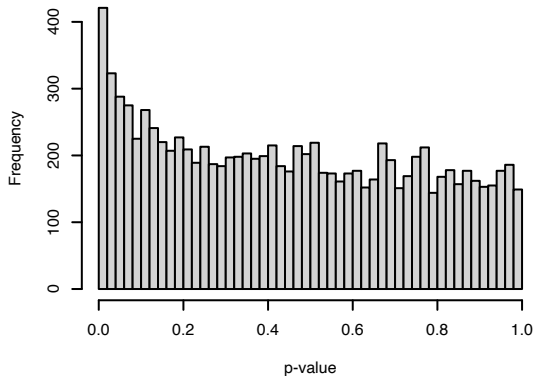
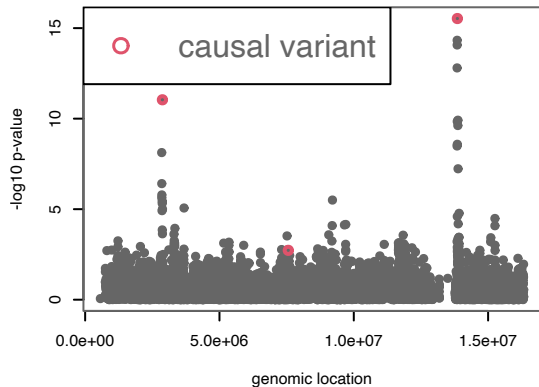
$$P(A \cap B) = P(A) \times P(B)$$

When we want to “control” two hypothesis testing....

$$P(\text{reject at least one true } H_0 \mid H_0) \leq \alpha + \alpha$$

What will be the α value that is guaranteed to control both?

Manhattan plot to show all the p-values...



We have $p = 10,000$ p-values. How do we decide which variants are significant?

The problem: we tested 10,000 hypotheses

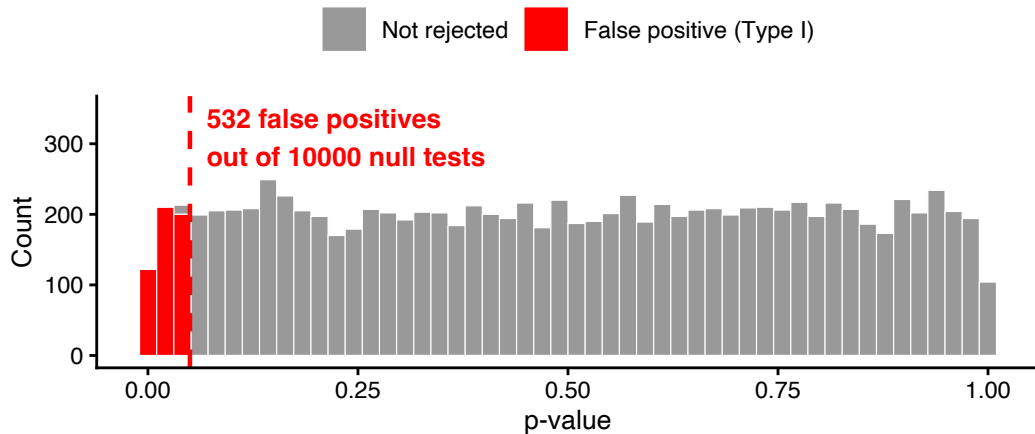
- We performed 10,000 simultaneous tests.
- At $\alpha = 0.05$, how many false positives do we expect **by chance alone**?

$$\mathbb{E}[\# \text{ false positives}] = 10,000 \times 0.05 = 500$$

- We need a principled way to control errors when testing many hypotheses at once.

Why does this matter for 10,000 tests?

All 10,000 tests are null — yet we get ~500 'significant' results



Conversely.... The dimensionality works unfavourably...

Each test at $\alpha = 0.05$: 5% chance of a false rejection if H_0 is true.

Variants (m)	$P(\geq 1 \text{ false rejection})$	
1	$1 - 0.95^1 = 0.05$	Just α
2	$1 - 0.95^2 = 0.098$	Nearly doubled
3	$1 - 0.95^3 = 0.143$	Nearly tripled
10	$1 - 0.95^{10} = 0.40$	40% chance!
100	$1 - 0.95^{100} = 0.994$	Almost certain
10,000	$1 - 0.95^{10000} \approx 1$	Guaranteed

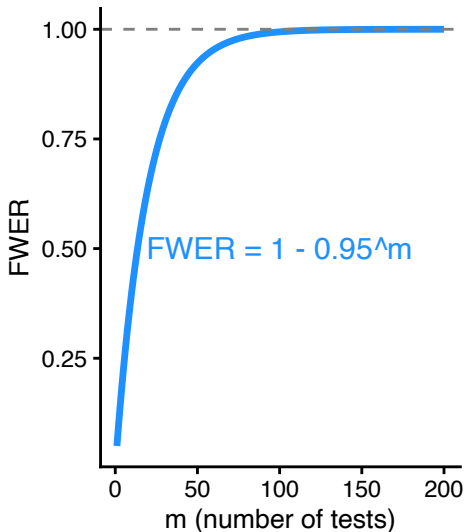
Family-wise error rate (FWER)

- $P(\text{incorrect} \mid H_0) = \alpha$; let $\alpha = 0.05$
- $P(\text{correct} \mid H_0) = 1 - \alpha = 0.95$
- For m **independent** tests, all under H_0 :

$$P(\text{all correct}) = (1 - \alpha)^m = 0.95^m$$

- **Family-wise error rate:**

$$\text{FWER} = P(\text{at least one error}) = 1 - 0.95^m$$



Bonferroni correction: controlling the FWER

By the **union bound**:

$$P(\text{at least one error}) \leq \sum_{i=1}^m P(\text{error on test } i)$$

Set each test's threshold to $\alpha_{\text{Bon}} = \frac{\alpha}{m}$:

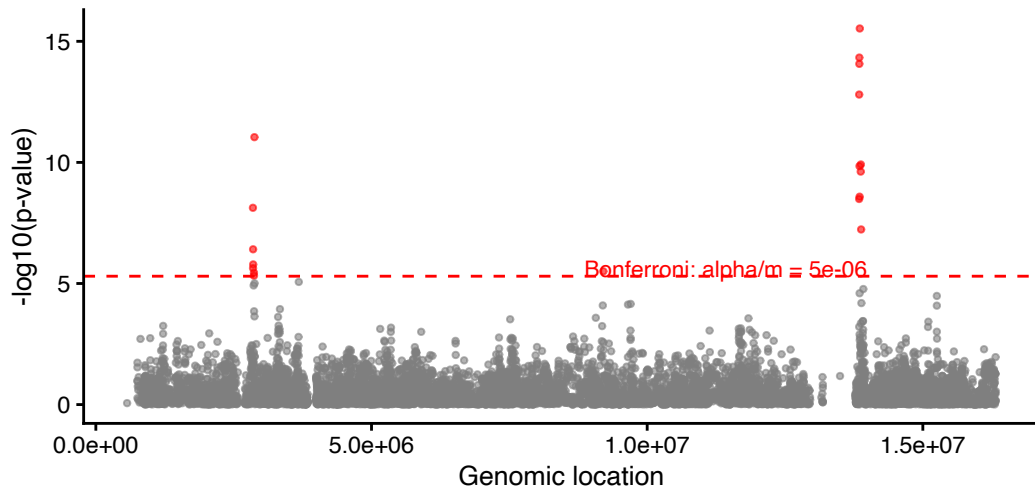
$$\sum_{i=1}^m \alpha_{\text{Bon}} = m \cdot \frac{\alpha}{m} = \alpha$$

Bonferroni correction: controlling the FWER

- Reject $H_{0,i}$ if $p_i < \alpha/m$, or equivalently,
- Adjust $p_{\text{Bon},i} = \min(m \cdot p_i, 1)$ and threshold at α .

Bonferroni on our GWAS example

Bonferroni: 18 rejections out of 10000 tests



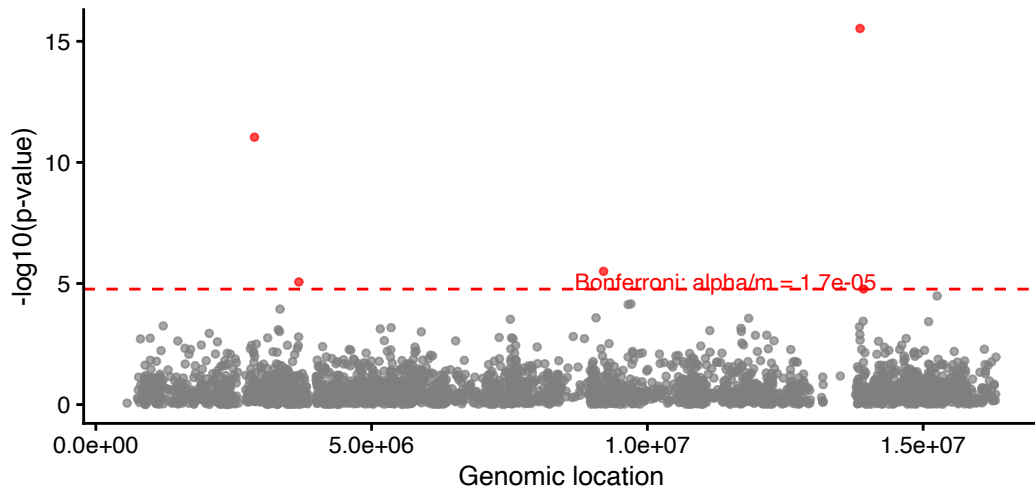
What if we remove correlated variants (LD clumping)?

After LD clumping, FWER methods work much better!

- **LD clumping** removes correlated variants, keeping only representative independent SNPs
 - It still takes some threshold parameter...
- Independent tests $\Rightarrow$ FWER and FDR are better aligned
- But: we lose power by discarding many variants

FWER control with LD clumping

After LD clumping: 5 rejections out of 2934 tests



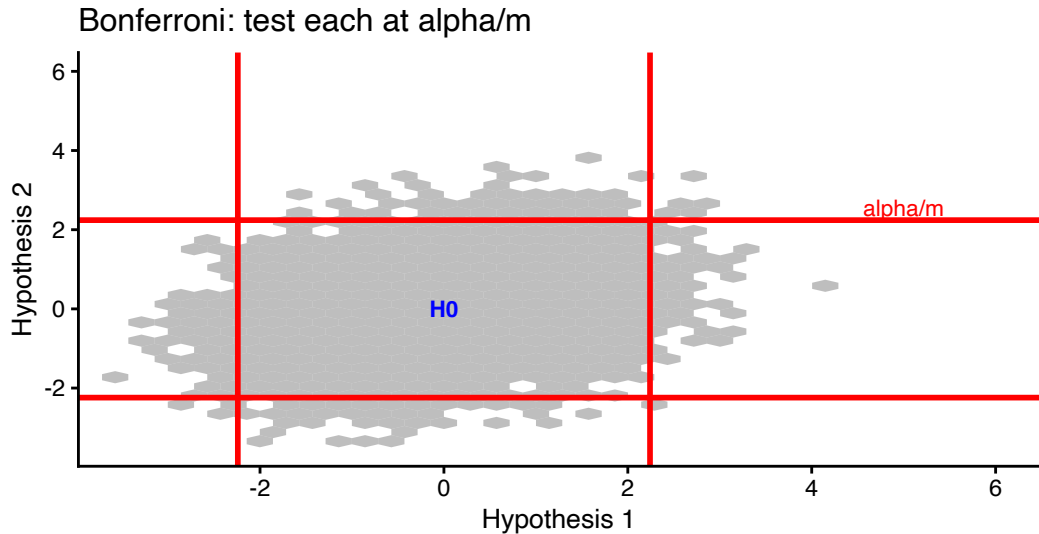
Holm–Bonferroni: a less conservative FWER control

- **Holm's step-down** procedure uses decreasing multipliers:

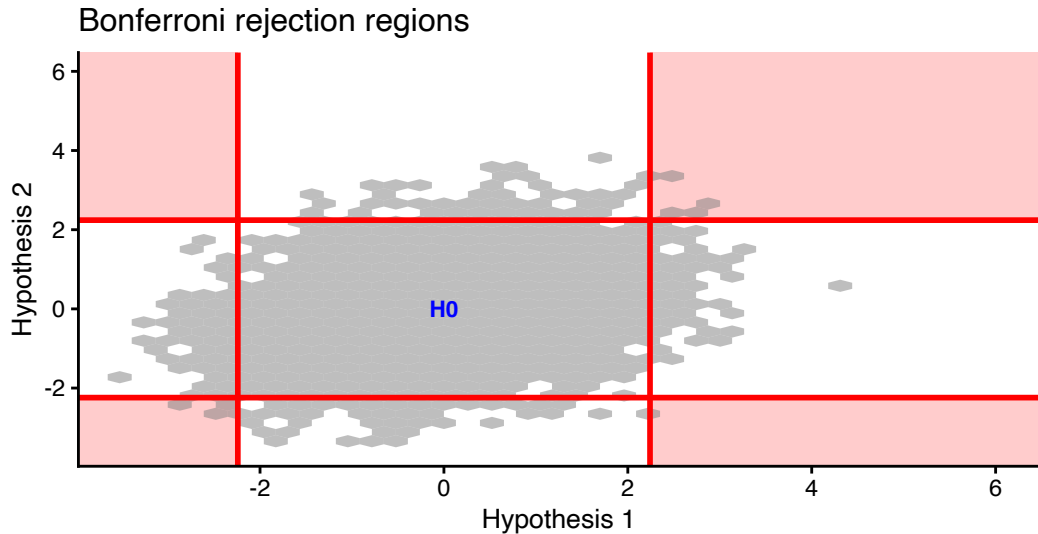
$$p_{\text{Holm}(1)} = m \cdot p_{(1)}, \quad p_{\text{Holm}(2)} = (m - 1) \cdot p_{(2)}, \quad \dots$$

- Still controls $\text{FWER} \leq \alpha$, but is **uniformly more powerful** than Bonferroni
- In R: `p.adjust(p, method = "holm")`

Bonferroni: uniform strict threshold

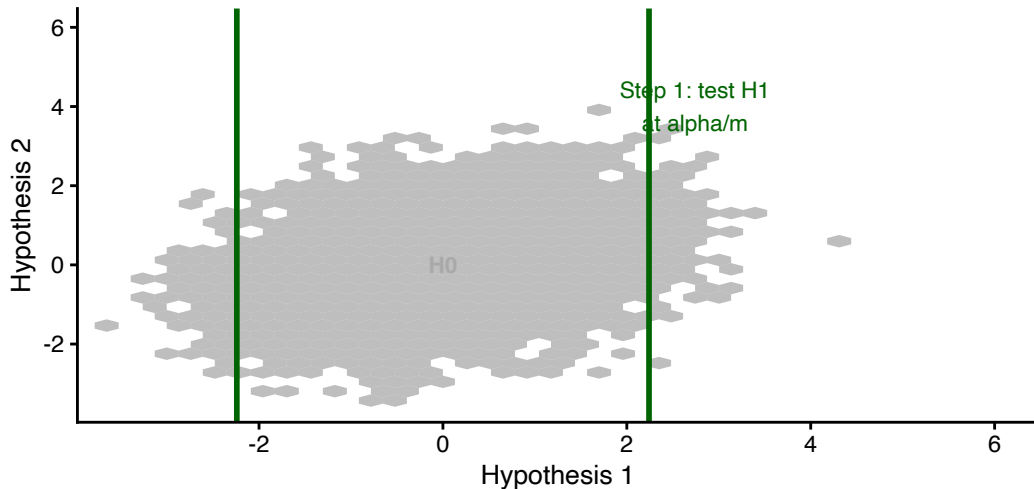


Bonferroni: uniform strict threshold

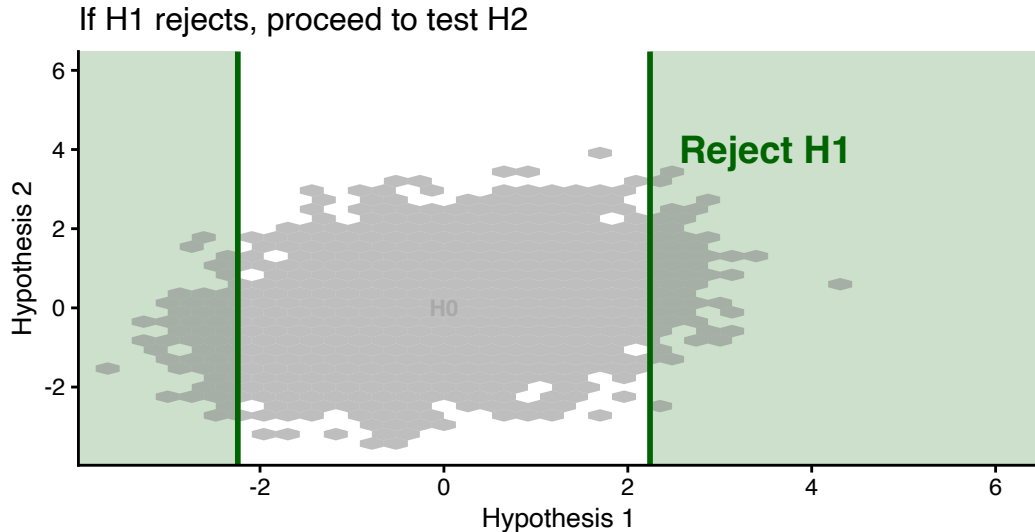


Holm: sequential testing, one axis at a time

Holm Step 1: test H1 (smallest p) at α/m

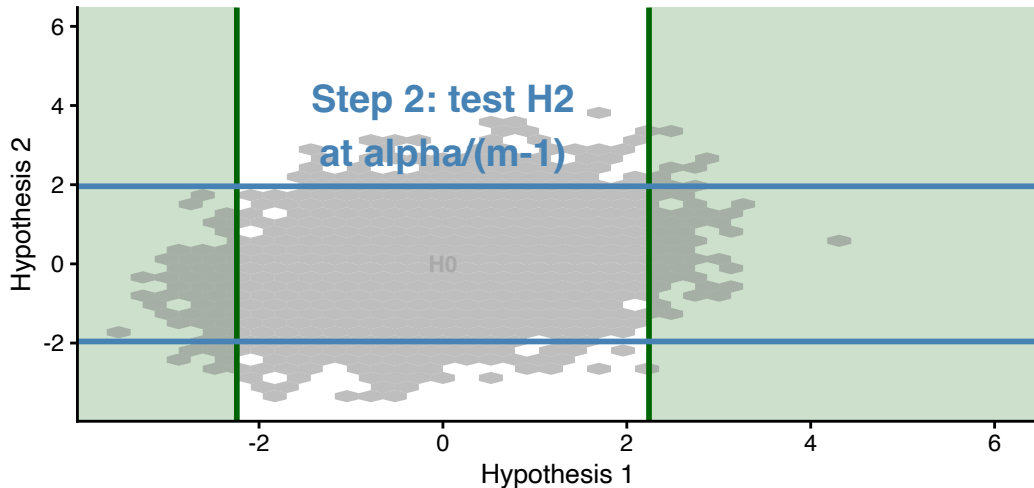


Holm: sequential testing, one axis at a time



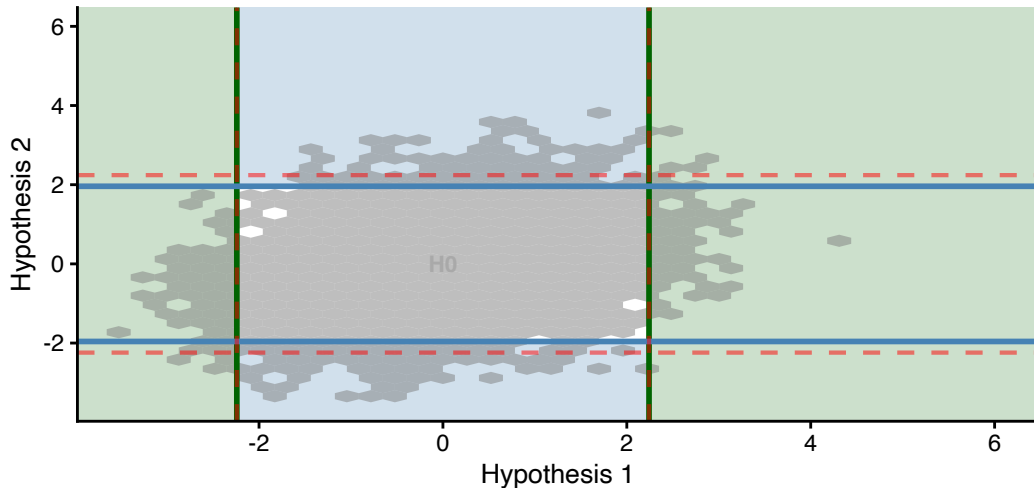
Holm: sequential testing, one axis at a time

Holm Step 2: test H2 at relaxed threshold $\alpha/(m-1)$



Holm: sequential testing, one axis at a time

Holm: sequential rejection regions (axis-by-axis)



Why is FWER impractical for genomics?

- So much about making **one single error** out of 20,000 tests?
- One easy way to guarantee no Type I errors: **reject nothing!**
 - But...

The trade-off

Being overly strict about Type I error $\Rightarrow$ greater Type II error.

- **Key idea:** in high-throughput biology, it is acceptable to make some errors, as long as we have a *comparatively* large number of true positives.

Pop-up quiz

Why do we use 5×10^{-8} as a GWAS significance threshold?

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The multiple testing table — not the probability

	Not rejected	Rejected	Total
H_0 true	U	V	m_0
H_0 false	T	S	m_1
Total	$m - R$	R	m

- R = total rejections (observed)
- V = false positives (unobserved)
- S = true positives (unobserved)

FDR vs. FPR vs. FWER

- **False Discovery Rate (FDR):**

$$\text{FDR} = \mathbb{E} \left[\frac{V}{R} \right] = \mathbb{E} \left[\frac{\text{\#false discoveries}}{\text{\#rejections}} \right] \quad (\text{define as 0 when } R = 0)$$

- **False Positive Rate (FPR):**
- **Family-Wise Error Rate (FWER):**

FDR vs. FPR vs. FWER

- **False Discovery Rate (FDR):**
- **False Positive Rate (FPR):**

$$\text{FPR} = \mathbb{E} \left[\frac{V}{m_0} \right] = \mathbb{E} \left[\frac{\text{\#false discoveries}}{\text{true null}} \right]$$

- **Family-Wise Error Rate (FWER):**

FDR vs. FPR vs. FWER

- **False Discovery Rate (FDR):**
- **False Positive Rate (FPR):**
- **Family-Wise Error Rate (FWER):**

$$\text{FWER} = P(V \geq 1) = P(\text{at least 1 false discovery})$$

Benjamini–Hochberg (BH) procedure

Control FDR at level q (e.g. $q = 0.05$):

- 1 Order the raw p-values: $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$
- 2 Find the largest rank j such that $p_{(j)} \leq \frac{j \cdot q}{m}$
- 3 Reject all hypotheses with rank $\leq j$

Adjusted p-values

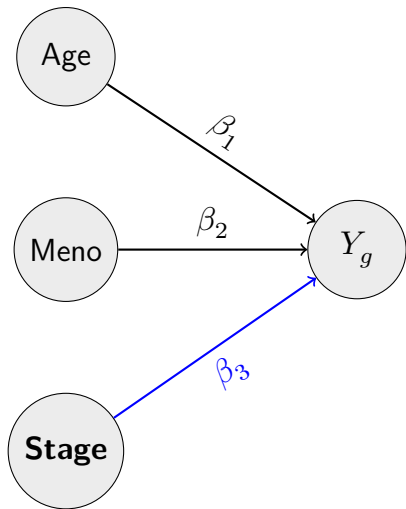
$$p_{(i)}^{\text{BH}} = \min_{j \geq i} \left(\frac{m}{j} p_{(j)} \right), \quad \text{capped at 1}$$

BH worked example ($m = 10, q = 0.05$)

Rank j	P-value	BH threshold $\frac{jq}{m}$	Adjusted p-value p^{BH}	Reject?
1	0.0008	0.005	0.008	✓
2	0.009	0.010	0.045	✓
3	0.127	0.015	0.423	
4	0.205	0.020	0.513	
5	0.396	0.025	0.750	
6	0.450	0.030	0.750	
7	0.641	0.035	0.916	
8	0.781	0.040	0.976	
9	0.900	0.045	0.993	
10	0.993	0.050	0.993	

Bonferroni ($\alpha/m = 0.005$) would only reject rank 1.

DEG: expression $Y \sim$ cancer stage in TCGA BRCA



Linear model for gene g :

$$Y_{ig} = \beta_{1g} \cdot \text{Age}_i + \beta_{2g} \cdot \text{Meno}_i + \beta_{3g} \cdot \text{Stage}_i + \epsilon_{ig}$$

where $\epsilon_{ig} \sim N(0, \sigma_g^2)$

Null hypothesis for gene g :

$$H_{0g} : \beta_{3g} = 0$$

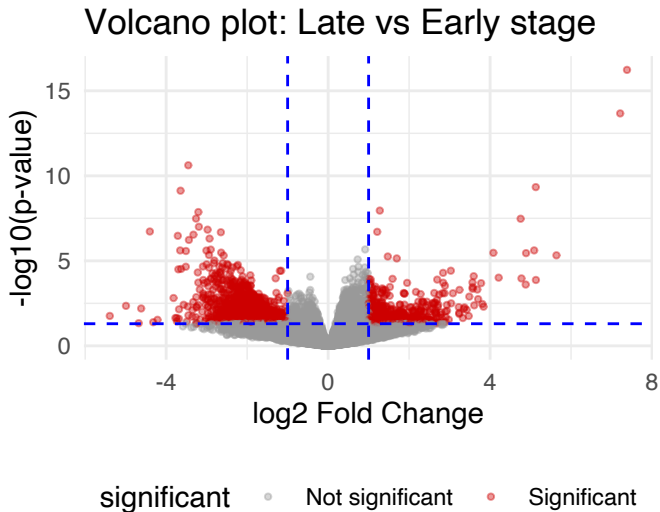
Testing $m = 20,000$ genes $\Rightarrow m = 20,000$ hypotheses!

Run limma-voom analysis

##		logFC	AveExpr	t	P.Value	adj.P.Val
##	ENSG00000221867.8	7.381752	-8.937562	8.505772	5.869385e-17	3.293606e-12
##	ENSG00000197172.10	7.213459	-9.570496	7.747667	2.130298e-14	5.977083e-10
##	ENSG00000233735.5	-3.453995	-9.963162	-6.750199	2.392598e-11	4.475355e-07
##	ENSG00000198930.12	5.128308	-7.603639	6.289275	4.605533e-10	6.460987e-06
##	ENSG00000233473.2	-3.645117	-9.021456	-6.211074	7.468179e-10	8.381537e-06

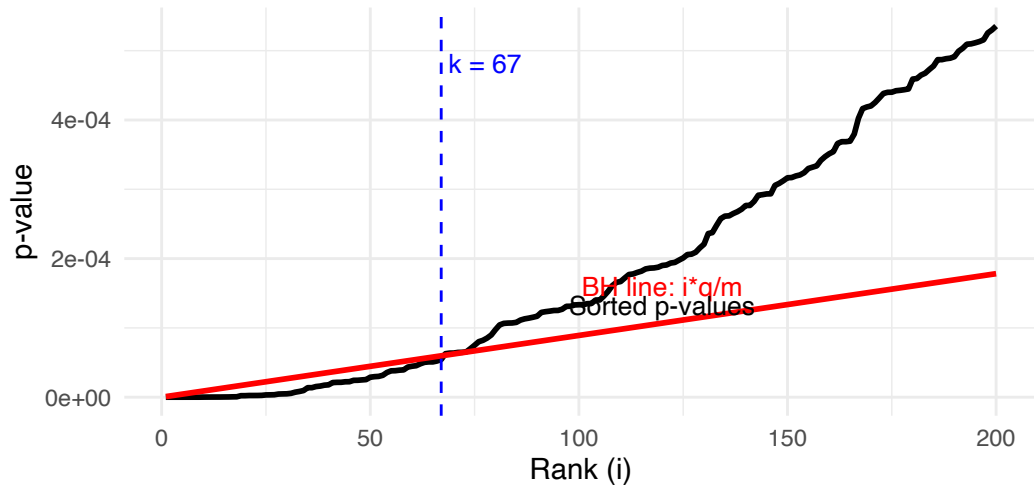
How did we get the “adjusted” p-value?

Volcano plot: visualizing effect size and significance



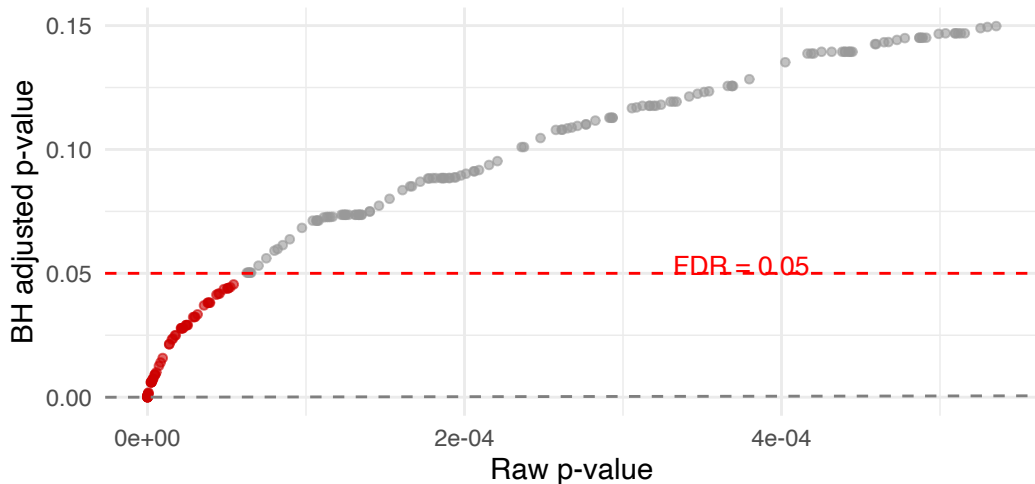
Benjamini-Hochberg procedure visualized

Reject 67 hypotheses at $FDR < 0.05$



Benjamini-Hochberg procedure visualized

67 genes significant at $FDR < 0.05$



Recap: what are we controlling?

FWER (Bonferroni, Holm): Controls **probability** of making **any** false discovery

$$\text{FWER} = P(V \geq 1) \leq \alpha$$

FDR (Benjamini-Hochberg): Controls **expectation** of **proportion** of false discoveries

$$\text{FDR} = E \left[\frac{V}{R} \right] \leq q$$

Key difference

BH controls the **expected** FDR, not the probability. In any single experiment, the actual FDR can be higher or lower than q .

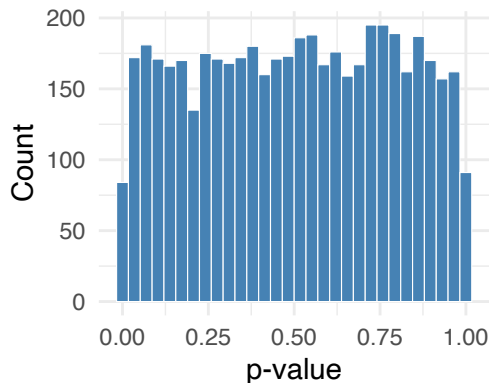
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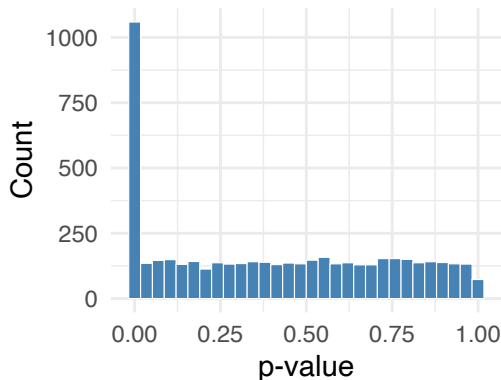
Well-behaved p-values: uniform under the null

All multiple testing methods assume that p-values under H_0 are **uniformly distributed**.

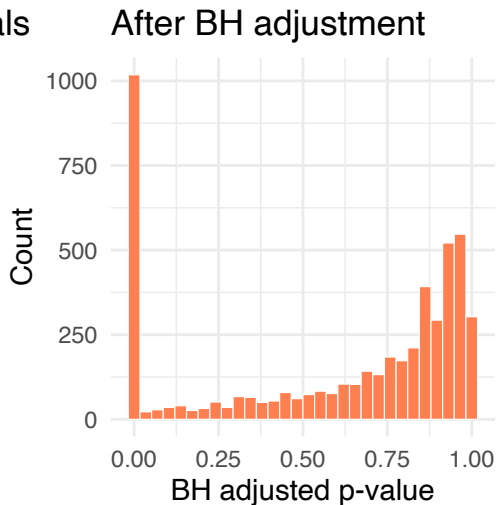
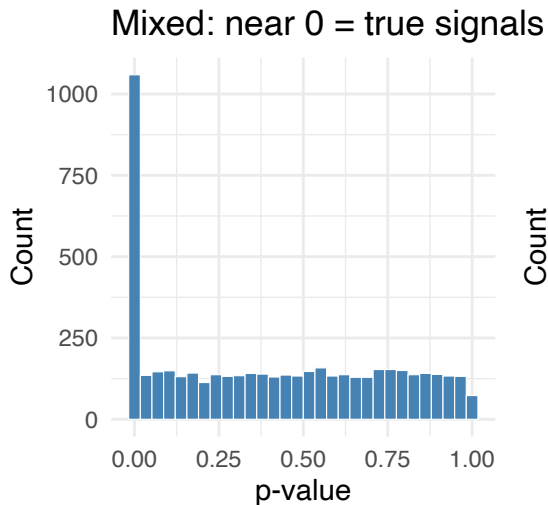
All null: uniform distribution



Mixed: near 0 = true signals



BH adjusted p-value distribution



Poorly behaved p-values

When p-values **misbehave**, FDR estimates become invalid.

Common causes:

- Model misspecification (wrong distributional assumptions)
- Uncontrolled variation or batch effects
- Selection bias / data filtering after testing

Great reference:

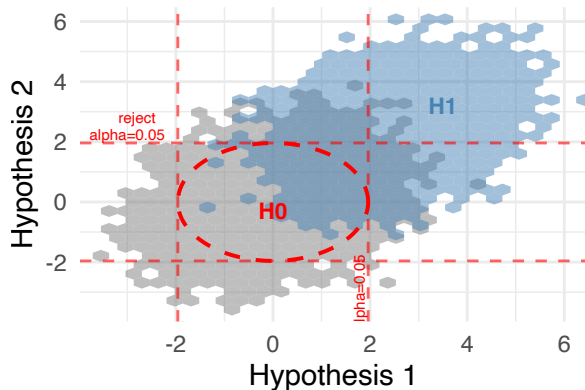
<http://varianceexplained.org/statistics/interpreting-pvalue-histogram/>

What can we do to make sure that p-values

Possible solutions:

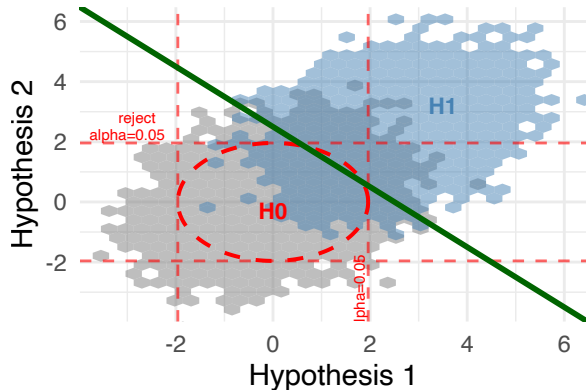
- **Nonparametric tests** (but limited flexibility for covariates)
- **Permutation testing**: shuffle group labels to construct an empirical null distribution; compare observed statistics to the permutation distribution
 - Limitation: computationally intensive; not ideal for very small sample sizes

What can we estimate from data?



- Can we estimate the distribution under H_0 ?
- Can we estimate the distribution under H_1 ?
- Do we need to estimate what's going on under H_1 ?

What can we estimate from data?



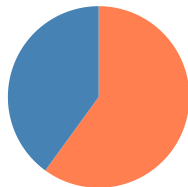
- Can we estimate the distribution under H_0 ?
- Can we estimate the distribution under H_1 ?
- Do we need to estimate what's going on under H_1 ?

Can we come up with an *optimal* boundary?

Bayesian view of multiple testing

Mixture model framework: Each hypothesis is either

- **Null** (H_0) with prior probability π_0
- **Alternative** (H_1) with probability $1 - \pi_0$



Status ■ Alternative ■ Null

Bayesian interpretation (note what's in the condition)
[@Storey2003-yd]:

$$\alpha(n) \approx P(H_1 \mid n < n^*)$$

Multiple Hypothesis Testing (in DEG, GWAS)

Bayesian view of multiple testing

We want to control: $P(H_0 \mid \text{reject}) = P(H_0 \mid p \leq t)$

Bayes rule:

$$P(H_0 \mid p \leq t) = \frac{P(p \leq t \mid H_0) \cdot P(H_0)}{P(p \leq t)}$$

- It is called “positive” FDR because of the denominator.
- What is the first term of the numerator?

[@Storey2003-yd]

Storey's q-value

Key insight: BH assumes **all** hypotheses are null ($\pi_0 = 1$)

But in reality, many hypotheses are **truly alternative** (non-null)

Storey's q-value: Estimates π_0 (proportion of true nulls) from data

Compare to BH:

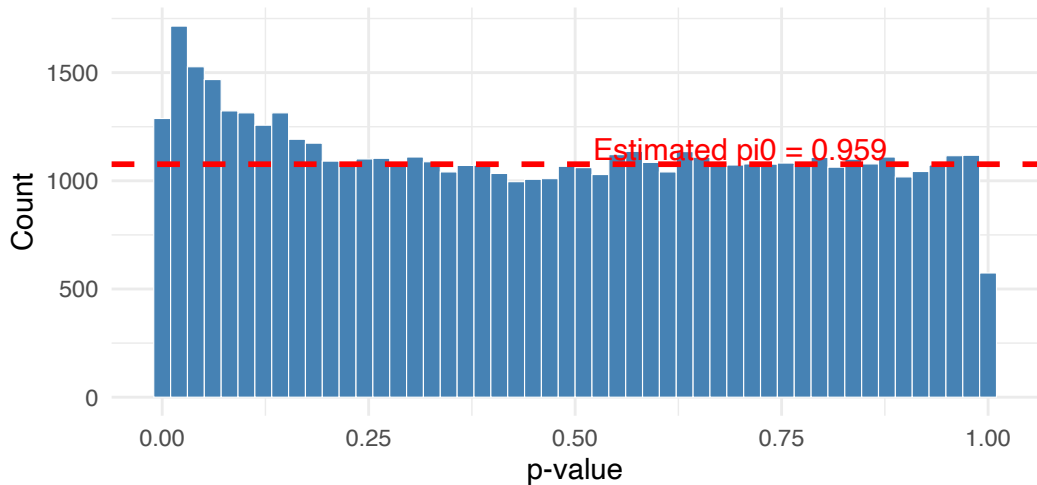
$$q(p_i) = \min_{j:p_j \geq p_i} \frac{\hat{\pi}_0 \cdot m \cdot p_j}{j}$$

$$\text{BH}(p_i) = \min_{j:p_j \geq p_i} \frac{m \cdot p_j}{j}$$

When $\hat{\pi}_0 < 1$: q-value is **less conservative** than BH

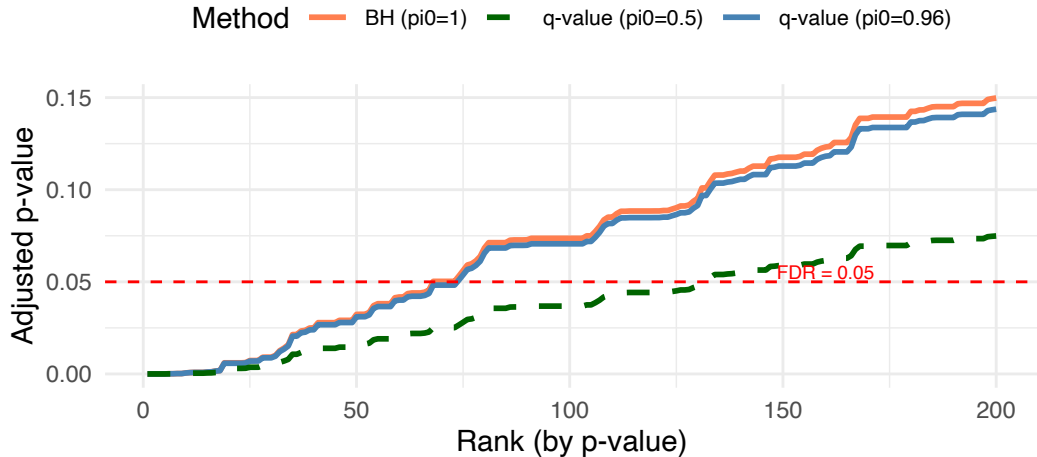
Estimating π_0 from p-value histogram

P-value histogram: red line estimates null density

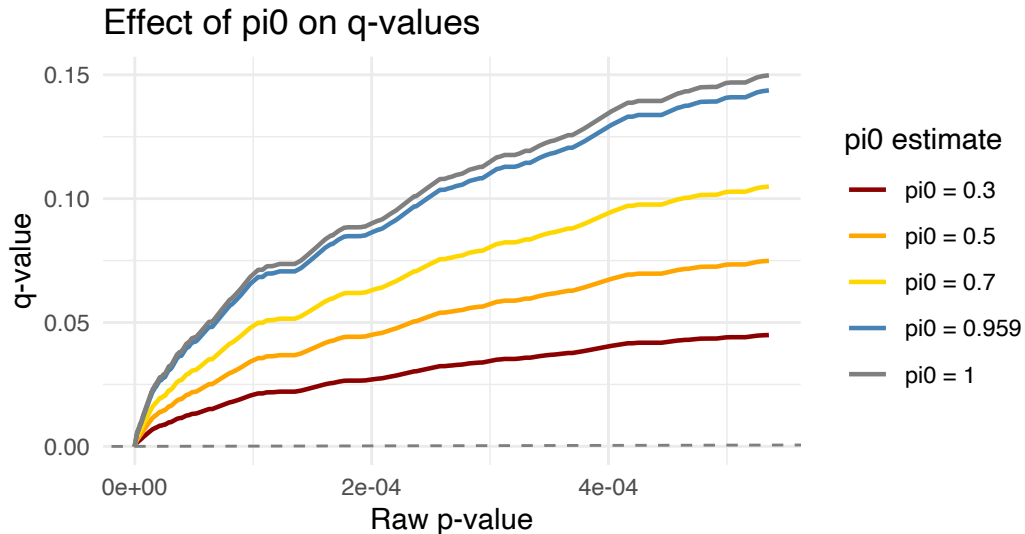


Comparing BH vs. q-value

BH vs q-value with different π_0



Effect of different π_0 estimates



When to use Storey's q-value vs. BH?

BH procedure:

- Conservative (assumes $\pi_0 = 1$)
- No assumptions about null proportion
- Valid (if independent)
- Recommended as default

Storey's q-value:

- Less conservative when $\pi_0 < 1$
- Estimates π_0 from data
- More powerful when many true signals
- Risk: π_0 estimation can be unstable with few tests

Rule of thumb: Use BH by default; consider q-value when you have > 1000 tests with many expected signals

Today's lecture

- 1 Hypothesis testing in terms of two errors
- 2 Controlling Family-Wise Error Rate
- 3 False Discovery Rate
- 4 FWER, FDR, then what?
- 5 Beyond global FDR: gene-by-gene localized control

Local false discovery rate (Efron 2001)

FDR: Expected proportion of false discoveries **among all rejections**

Local fdr: Probability that **a specific test** is null

$$\text{lfdr}(z_i) = P(H_0 \mid z_i) = \frac{\pi_0 f_0(z_i)}{f(z_i)}$$

Numerator: $\pi_0 f_0(z_i)$

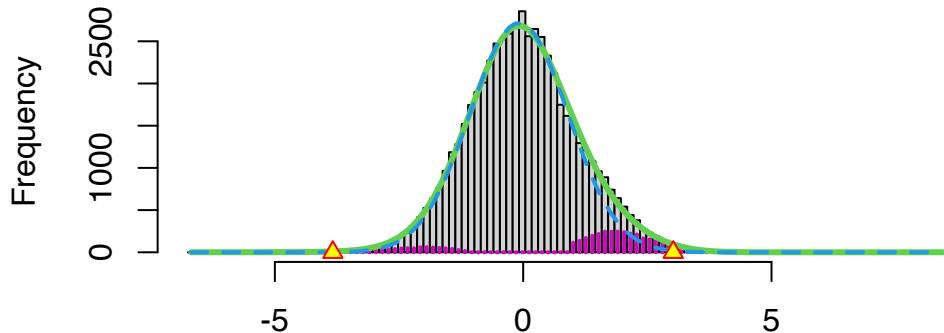
- Null contribution at z_i

Denominator: $f(z_i)$

- Total density at z_i

Can estimate from empirical distribution of z-scores

Local fdr: DEG example



MLE: delta: -0.085 sigma: 0.985 p0: 0.932

CME: delta: -0.074 sigma: 1.02 p0: 0.954

q-value vs. local fdr

q-value (Storey)

$$q(p_i) = \min_{j: p_j \geq p_i} \frac{\pi_0 \cdot m \cdot p_j}{j}$$

- Based on **p-values**
- Ranks tests by p-value
- Minimum FDR at threshold p_i
- **Tail probability** interpretation

Local fdr (Efron)

$$\text{lfdr}(z_i) = \frac{\pi_0 f_0(z_i)}{f(z_i)}$$

- Based on **test statistics** (z-scores)
- Density ratio at specific z_i
- Posterior $P(\text{null} \mid z_i)$
- **Point-wise** interpretation

Both use π_0 , but q-value uses **cumulative** p-values; local fdr uses **density** at each point

What FDR doesn't tell us

FDR controls the **expected proportion** of false discoveries

But for each gene, we want to know:

- Is this gene **truly differentially expressed**?
- What is the **direction** of the effect?
- What is the **magnitude** of the effect?

The problem

A small p-value tells us there's an effect, but not its direction or size with certainty

Local False Sign Rate (lfsr)

Local fsr: Probability that the **sign** of an effect is wrong

$$\text{lfsr} = P(\text{sign}(\beta) \neq \text{sign}(\hat{\beta}) \mid \text{data})$$

Key idea: Account for uncertainty in **both**:

- Whether there's an effect ($\beta \neq 0$)
- The direction of the effect (sign of β)

Adaptive Shrinkage (ashr)

ashr (Stephens 2017): Empirical Bayes approach

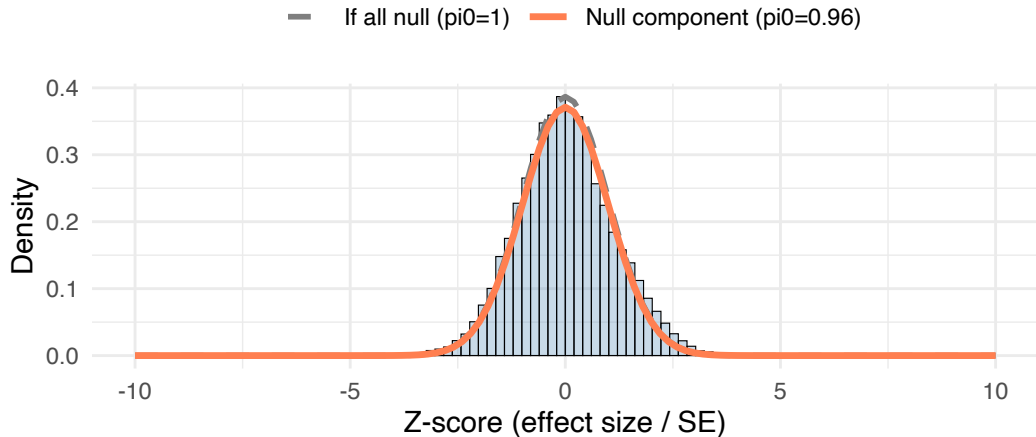
Key features:

- ① Estimates effect size distribution from data
- ② Shrinks estimates toward zero (like in ridge regression)
- ③ Provides posterior probabilities for sign and magnitude
- ④ Computes local false sign rate (lfsr)

More conservative than FDR for **directional claims**

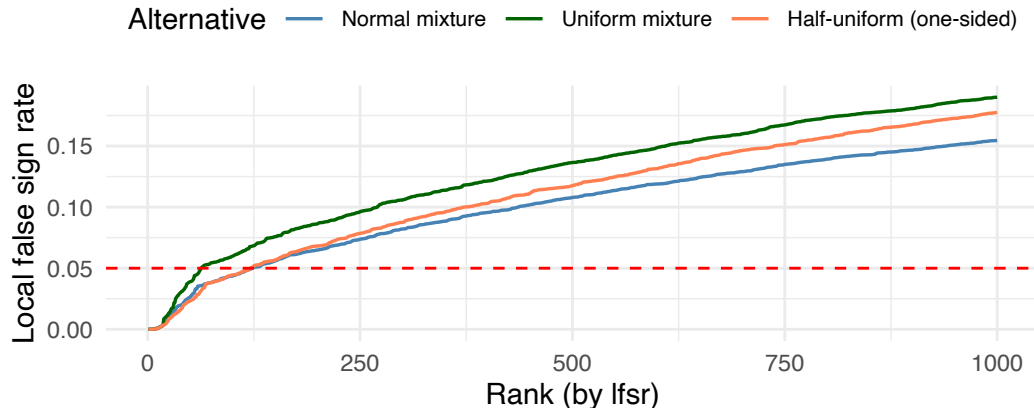
ashr: Estimating null vs alternative distributions

Observed vs null distribution



ashr: Different alternative distributions

Effect of alternative distribution assumption



Note: qvalue estimates $\pi_0=0.96$ from p-values; ashR can give different estimates from effect sizes

Why shrinkage?

Problem: Raw effect size estimates are **noisy**

Genes with large SE: - Less precise measurements - Estimates unreliable - May be spuriously large

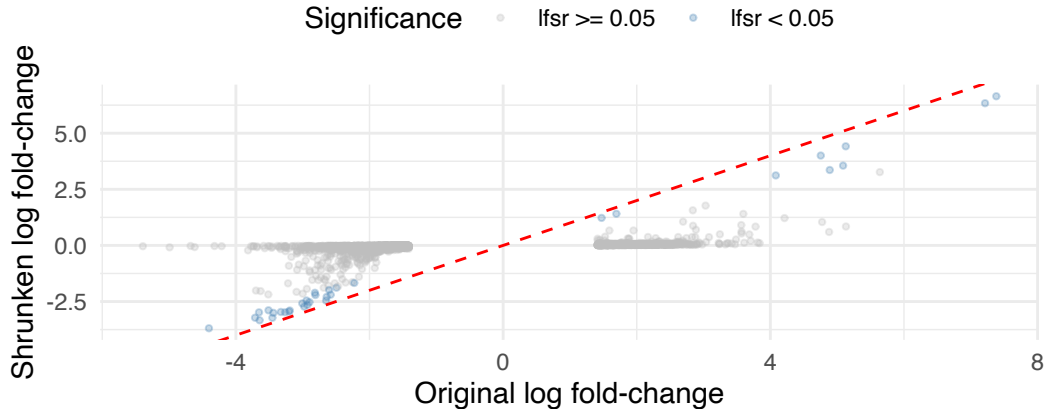
Genes with small SE: - More precise measurements - Estimates more reliable - Should be trusted more

Solution: Shrink uncertain estimates toward zero

- Small/noisy effects $\rightarrow$ shrunk more
- Large/precise effects $\rightarrow$ preserved
- **Borrows strength** across all genes

ashr in action: Effect size shrinkage

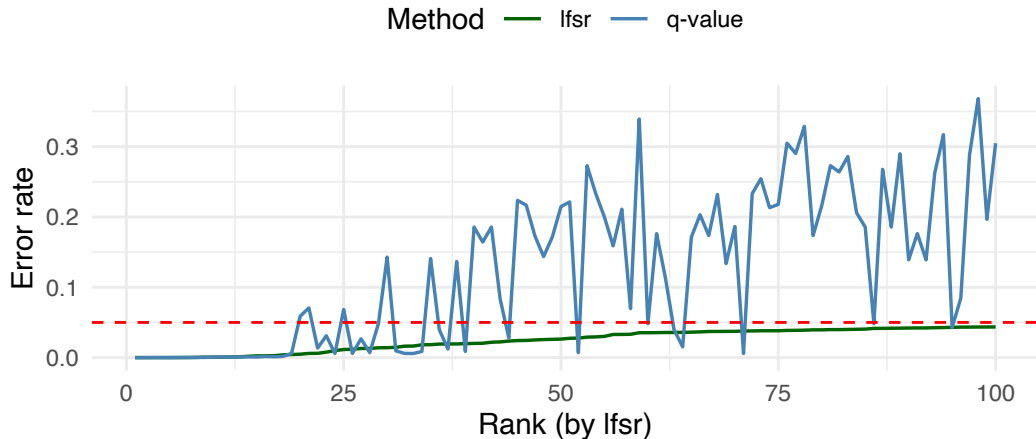
Effect size shrinkage by ash



Points below diagonal: shrunk toward zero (less confident). Points near diagonal: preserved (more confident).

ashr: local false sign rate vs q-value

Local false sign rate vs q-value



Summary: Multiple testing methods

Control FWER (strict):

- **Bonferroni:** α/m
 - Simple, conservative
 - Use for small m or independent tests
- **Holm:** Sequential Bonferroni
 - More powerful than Bonferroni
 - Still controls FWER

Control FDR (flexible):

- **Benjamini-Hochberg:** Assumes $\pi_0 = 1$
 - Standard FDR control
 - More discoveries than FWER
- **Storey q-value:** Estimates π_0
 - More powerful when $\pi_0 < 1$
 - Uses p-values only
- **ashr:** Uses effect sizes + SE
 - Accounts for measurement precision
 - `lfsr` for directions